COE 202 – F01

College of Engineering & Computer Science

Computer Engineering Department

A detector for measuring the CO2 emissions in the atmosphere

Digital Logic Design [COE-202]

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ABSTRACT

The aim of the project was to implement a detector that measures CO2 emissions and to provide real-time monitoring of the data. The successful completion of the final assembly project demonstrates the effectiveness of the detector in accurately measuring CO2 emissions. The project required a variety of material, including Arduino Nano, MG-135 sensor, OLED display.

INTRODUCTION

Carbon dioxide emissions, largely by-products of energy production and use, account for the largest share of greenhouse gases, which are associated with global warming. To stay informed about the percentage of emissions in the atmosphere and to take action to reduce the carbon footprint, our project is a detector system that continuously monitors CO2 levels in the environment and provides real-time data to help making decisions to reduce carbon footprint. The project has a signal to inform that the emissions exceeded 1000ppt.

OBJECTIVES

1.Design a detector system for carbon dioxide emissions in the atmosphere.

2.Raise awareness of human activities contributing to climate change and the resulting health impacts.

3.Design an affordable CO2 emission detector system in low-resource environments to help people own it.

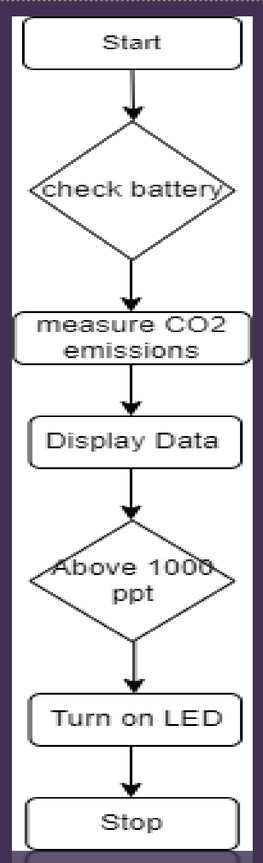

Figure 1. Flow Chart of The Project.

MATERIALS & METHODS

OLED Display: An OLED display acts as the visual output device in the circuit. It allows to display the CO2 concentration readings obtained from the sensor in a clear and user-friendly format.

MG-135 sensor: MG-135 is used to monitor and measure CO2 emissions in the atmosphere and provide data to display on a digital screen for real-time tracking.

Arduino IDE: Coding in Arduino IDE will specify how to Read data from the CO2 sensor, Interpret the sensor readings and convert them into meaningful CO2 concentration values.[

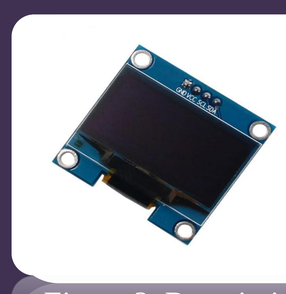

Figure 2. Description of Graph / Illustration

Figure 3. Description of Graph / Illustration

Figure 4. Description of Graph / Illustration

RESULTS & DISCUSSION

According to the data, Saudi Arabia's CO2 emissions have been steadily increasing over the years. This shows the importance of implementing a detector to measure CO2 emissions. Therefore, a decetor measure was implemented to measure CO2 emissions in the atmosphere to provide a real-time monitor of the emissions and to inform that the emissions exceeded 1000 ppt, providing a clear visual alert.

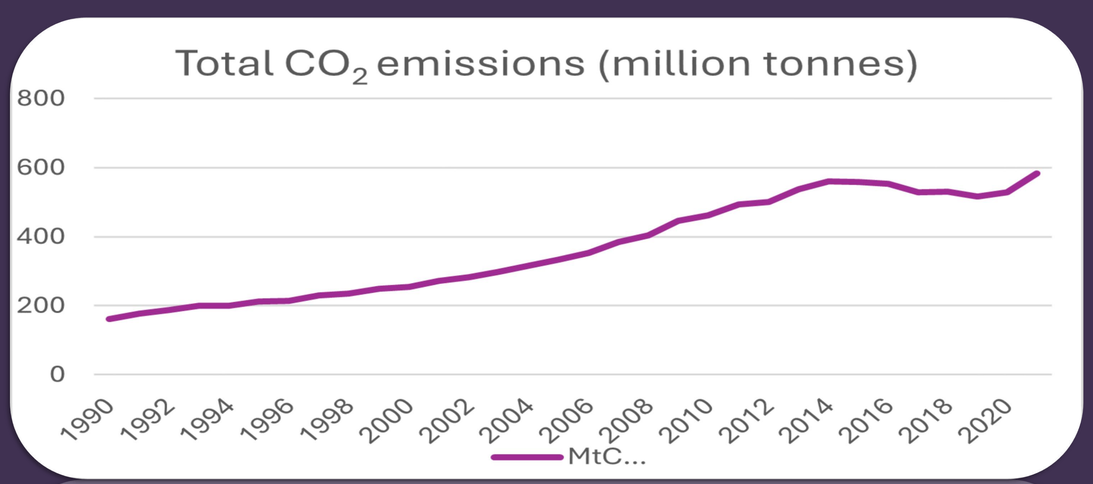

Figure 1. Carbon dioxide emissions in Saudi Arabia increased steadily over the first years

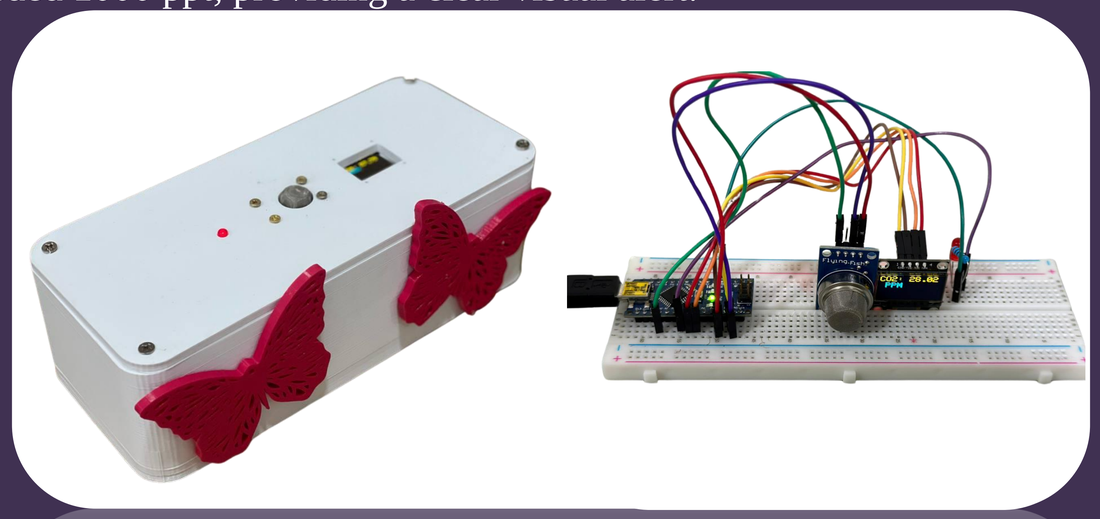

Figure 1. Description of Graph / Illustration

Figure 2. Description of Graph / Illustration

CONCLUSION

The aim of the project was to implement a detector that measures CO2 emissions and to provide real-time monitoring of the data. The successful completion of the final assembly project demonstrates the effectiveness of the detector in accurately measuring CO2 emissions.

REFERENCES

S. Al Mutairi, M. Al alwan, R. Al Roqeei .2024 .” A detector for measuring the CO2 emissions in the atmosphere“. Mustaqbal University, Saudi Arabia.